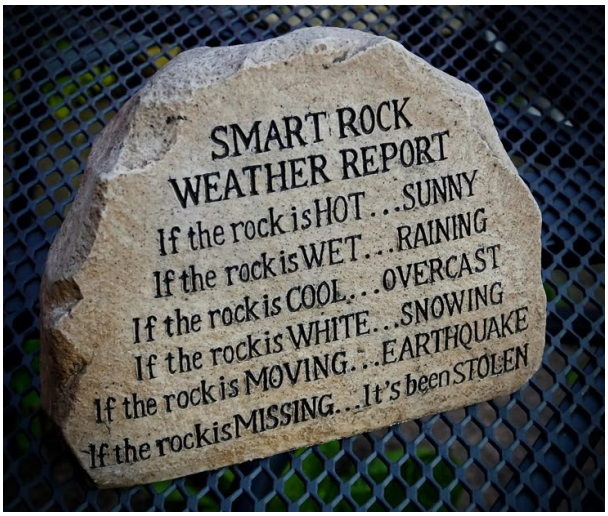
Science Olympiad
Meteorology B Test
Greenbrier Invitational



Team Name: _____

Team Number: _____

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Directions:

A. You will have 50 minutes to complete this test

B. Ties will be broken based on correctness of the following questions: 6, 7, 4, 1

C. Some questions are supposed to be difficult, so don't worry if you aren't able to finish all the questions.

Page:	2	3	4	5	6	Total
Points:	9	10	11	8	8	46
Score:						

1. (2 points) Order the following layers of the atmosphere from lowest to highest altitude:
   - Mesosphere
   - Troposphere
   - Exosphere
   - Thermosphere
   - Stratosphere

2. (2 points) Which layer of the atmosphere is the most dense, and why?

3. (3 points) For each layer of the atmosphere, what is the most common pollutant?

4. (2 points) How much does the sun's longitude change with respect to the equator each day?

5. (2 points) Using your answer from the previous question, what would the longitude of the subsolar spot be on March 30th? Show your work.

For question 6, write the appropriate instrument to monitor the atmospheric condition. There may be more than one correct answer, but there will be only one best answer.

6. (6 points) Air temperature, atmospheric pressure, relative humidity, wind speed, wind direction, precipitation amount, cloud base height, solar radiation intensity, atmospheric ozone concentration, particulate matter, vertical temperature profile, and atmospheric water vapor content.

7. (2 points) Define and explain the meteorological significance of a wet-bulb temperature

8. (3 points) At the wet-bulb temperatures $0°C, 10°C$, and $20°C$, what cloud formations would be expected

9. (2 points) What is the most important meteorological condition in order for a green flash to happen, and why?

10. (2 points) Compare and contrast mercury-based and alcohol-based thermometers based on their effectiveness at different temperatures

11. (2 points) Which region of the world has the most virga, and why?

12. (2 points) If the Coriolis effect magically disappeared immediately, what would be expected over the course of the next five minutes?

For questions 13-16, use the following information: On a summer afternoon, a weak surface low and a nearby stationary front allow warm, moist air to pool over a flat inland region. Daytime solar heating increases instability, producing scattered cumulus that grow into isolated thunderstorms, but weak wind shear prevents storm organization. The storms produce brief heavy rain and occasional lightning, with no hail or damaging winds.

13. (1 point) Identify the lifting mechanism responsible for thunderstorm initiation in this scenario.

14. (2 points) Explain why the thunderstorms remain nonsevere despite the presence of warm, moist air.

15. (2 points) Describe how the vertical wind shear profile influences storm type and lifetime in this situation.

16. (3 points) If mid-level dry air were introduced into this environment, analyze how precipitation type and downdraft strength might change without necessarily producing severe storms.

Use the following information to answer questions 17-20: As the single-cell thunderstorms develop, they produce brief, localized heavy rain and occasional lightning strikes, but most precipitation evaporates before reaching the ground in some areas. Gusty downdrafts create temporary wind shifts at the surface, and minor puddling occurs in low-lying areas, but no flooding develops. The storms quickly dissipate as the updraft collapses, leaving behind scattered clouds and a temporary drop in temperature.

17. (1 point) Identify what meteorological phenomenon is observed when precipitation evaporates before reaching the ground.

18. (2 points) Explain why surface gusts occur during these short-lived thunderstorms.

19. (2 points) Describe how uneven rainfall distribution in this scenario could impact local areas differently.

20. (3 points) Analyze how the combination of virga and downdrafts affects near-surface temperature and humidity immediately after the storm.